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قسم: العلاج التحفظي

خطة المقترح البحثي لنيل درجة ماجستير العلاج التحفظي

تحت عنوان

باللغه العربيه:

دراسة مقارنة للتأكل و خشونة سطح الراتينج المركب القابل للحقن و التقليدي ـ دراسة في المختبر

باللغه الانجليزيه:

Comparative Study of Wear Resistance and Surface Roughness of Inject able Versus Conventional Composite Resin - In Vitro Study

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English Title: COMPARATIVE STUDY OF WEAR RESISTANCE AND SURFACE ROUGHNESS OF INJECTABLE VERSUS CONVENTIONAL COMPOSITE RESIN - IN VITRO STUDY

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Keywords: Surface roughness, wear, injectable composite, conventional composite.

Supervision Committee

1. **Prof. Wegdan M. Abdel-Fattah**
2. **Asst. Prof. Emad M. El-Sayed (Main supervisor)**

Role of Supervisors

1. Prof. Wegdan M. Abdel-Fattah

Conceptualization/formulation of study plan and design of the study.

Analysis and interpretation of study results.

Supervision of study execution and writing.

Guidance of paper/thesis writing

2. Asst. Prof. Emad M. El-Sayed

Supervision of the clinical/experimental procedures

Conceptualization/formulation of study plan and design of the study.

Supervision of study execution and writing.

Revising the thesis and/or manuscript

CONTENTS

Contents	Page

ABSTRACT

Background: Dental composite resins utilized as anterior restorations must have mechanical properties that can withstand the chewing and abrasion forces encountered in the oral cavity, in order to maintain esthetics and function.

Aim: The aim of this in vitro study is to compare the wear and surface roughness of novel highly-filled flowable (Injectable) and conventional composite resin restorative materials.

Materials and Methods: A set of thirty six disc-shaped specimens (n=36) measuring 10mm in diameter and 1mm in depth will be prepared for the study by using a teflon mold. These specimens will be divided into 3 groups. Group 1; A nano-hybrid conventional composite Shofu Beautifil II LS. Group 2; Highly filled flowable (Injectable) nano-hybrid composite (GC G-aenial Universal Injectable) and Group 3; Highly filled flowable (Injectable) nano-hybrid composite (Beautifil Flow Plus X 00). The specimens will be subjected to 10,000 cycles of brushing using a custom made artificial tooth brushing machine. The amount of wear will be determined by weighing each specimen before and after the tooth brushing test, using an electronic sensitive balance (RADWAG Wagi Electroniczne, AS 220-R2). Average surface roughness (Ra) will be evaluated before and after the tooth brushing test, using contact profilometer (MarSurf PS10).

Analysis: Data will be collected, tabulated and statistically analyzed.

Keywords: Surface roughness, wear, injectable composite, conventional composite.

INTRODUCTION

Composite resins are now the preferred material for direct restorations due to their excellent aesthetic qualities, enhanced adhesive capabilities and tooth structure presentations.[1,2]

One of the drawbacks to anterior composite is its technique sensitivity. This is where highly-filled flowable, named ‘injectable’, composites come into play, with a viscosity low enough to be injected as one increment to simplify the method of application.

Historically, flowable composite resins were not recommended for high-stress-bearing areas or larger cavities, however, the newly designed highly filled, nano-hybrid flowable composite resins could possibly exhibit better wear characteristics than the widely used conventional nanohybrid resin composites. This is due to some novel changes in the filler size and composition and filler silanisation.[1]

Clinically, the wear of anterior composite restorations mainly occurs as a result of continuous toothbrushing.[3]

There are two types of wear: two-body wear and three-body wear. The difference between both types is the presence of abrasive particles (toothpaste) between the two functional surfaces.

The wear resistance in composite resin restorations is mainly affected by the balance between the nature of inorganic fillers and the type of organic matrix. Also, the weak bond between organic and inorganic fillers may lead to the detachment of fillers and increased surface roughness.[4]

The surface roughness of composite resin largely depends on the size, shape, and percentage of inorganic filler particles. The smaller the filler particle sizes and the greater the percentage by volume, the more superior the physical and mechanical properties.[5]

Increased surface roughness may cause discoloration, loss of gloss, staining, and a higher retention of plaque. It has been shown that achieving an average surface roughness (*R*a) value below 0.2 μm is important in order to reduce the likelihood of surface plaque accumulation, caries development, and periodontal inflammation.[5]

The average surface roughness (Ra) of a dental composite resin can be measured using a contact profilometer, non-contact optical profilometry, or atomic force microscopy (AFM). These techniques provide precise assessments of surface irregularities, with Ra values typically expressed in micrometers (µm). It will be preferred to use contact profilometry due to the ease of process, availability, accuracy and ability to give a quantitative result.[6]

Some highly filled flowable (injectable) composites have been shown to have inferior mechanical properties to conventional composites,[7] whereas others have been shown to perform the same as, if not better than, conventional composites.[8] Therefore, there is a need to evaluate some highly filled 'injectable' composites currently on the market and to investigate their true functional performance.

Accordingly the aim of this study is to compare the surface roughness and wear resistance characteristics of two highly filled, nano-hybrid injectable composite resin materials (G-aenial Universal Injectable and Beautifil Flow Plus X F00) versus conventional nano-hybrid composite resin (Shofu Beautifil II LS).

The null hypothesis is that there would be no statistically significant difference in the wear resistance and surface roughness values between these forms of dental composite resin materials (injectable and conventional).

AIM OF THE STUDY

General Aim

The aim of this study is to evaluate two mechanical properties of two available highly filled flowable (injectable) nano-hybrid composites.

Specific Aim

The aim of this study is to compare wear resistance and surface roughness of injectable versus conventional composite resin.

MATERIALS AND METHODS

This study will be an in-vitro laboratory study.

Study Setting and Location

The sample preparation will be conducted in the laboratory of the Faculty of Dentistry, Pharos University in Alexandria, Egypt.

The experimental procedure will be conducted in the Conservative Dentistry Department laboratory of the Faculty of Dentistry, Alexandria University.

Study Sample and Sample Size Estimation

The study was designed to be powered to detect the wear resistance and surface roughness of injectable composites. The minimal sample size is calculated based on a previous study aimed to investigate and compare the wear and flexural strength of two highly filled (injectable), one flowable, and a conventional composite.(1) Authors concluded that highly filled flowable composite resins with nano filler particles outperformed a conventional flowable and a conventional composite resin in terms of wear resistance and flexural strength and may be suitable to use in occlusal, load-bearing areas.

The sample size was calculated to detect the difference in the wear resistance and surface roughness of novel, highly filled flowable (Injectable) and conventional form composite resin restorative materials. Results of a pervious study(1), adopting a power of 80% (B-0.20), and a level of significance 5% (a error accepted =0.05), the minimum required sample size was found to 10 specimens per group (number of subgroups-3) (Total sample size=30 specimens). Any specimen loss from the study sample due to processing error will be replaced to maintain the sample size.

Software

The sample size was calculated using GPower version 3.1.9.2(9)

Materials

The materials, composition and manufacturers used in the study are tabulated in table 1.

Table (1)

Material	Composition - Resin Matrix	Composition - Filler	Filler by Volume	Type	Manufacturer
Shofu Beautifil Flow Plus X F00 (Shade A2)	Bis-GMA TEGDMA	Multifunctional glass filler S-PRG filler	47%	Nanohybrid Injectable	Shofu Inc., Kyoto, Japan
G-aenial Universal Injectable (Shade A2)	Dimethacrylate monomers	Barium glass, Silica	50%	Nanohybrid Injectable	GC Corporation, Tokyo, Japan
Shofu Beautifil II LS(Shade A2)	Bis-GMA UDMA Bis-EMA PEGMA TEGDMA	Multifunctional glass filler S-PRG filler	68%	Nanohybrid Conventional	Shofu Inc., Kyoto, Japan

Equipment

1. Custom made wear brushing simulator machine.*
2. Blender CAD Software.**
3. Marsurf 10 Contact Profilometer.***
4. RADWAG Wagi Electroniczne, AS 220-R2 Electronic Sensitive Balance.****

* Dental Biomaterial Department, Faculty of Dentistry, Alexandria University
** Blender Institute B.V., Buikslotermeerplein 161, 1025 ET Amsterdam
*** Mahr GmbH, Carl-Mahr-Str. 1, D-37073 Goettingen
**** Warszawa, ul. Białołęcka 218A, 03-253 Warszawa

Intervention

Grouping

Group I (GC G-aenial Injectable Composite)

Group II (Shofu Flow Plus X 00)

Group III (Shofu Beautifil II LS)

Specimen Preparation

Thirty six composite disk specimens will be prepared using a 10 mm diameter and 1 mm depth CAD/CAM milled custom Teflon mold. The composite resin of each specimen will be packed into the mold spaces as one increment. Light curing will be performed using an LED light curing unit (MiniS, Woodpecker, Guilin, Guangxi, China) for 20 seconds with a light intensity of 800 mW/cm². Light curing will be done through a mylar strip underneath a transparent microscopic glass slide to achieve a flat surface on the composite disk, with the light cure device in direct contact with the microscope slide. To ensure complete polymerization of the composite disk, additional multidirectional curing will be performed (3 times for 20 seconds each). Finishing of the disk will be carried out using a no. 12 scalpel blade (Suzhou Kyuan Medical Co, Jiangsu, China).[10]

After separation of specimens carefully from the mold, the diameter of each disc will be measured using a digital caliper. The specimens will be immersed in distilled water in a closed glass container at 37°C for 24 hours.[11]

Each specimen will be removed from the glass container and dried using a sterile disposable sponge.

Weighing the Samples Before Starting Toothbrushing Test

Each sample will then be weighed using an electronic sensitive balance (RADWAG Wagi Electroniczne, AS 220-R2).

Surface Roughness Measurement before Starting Toothbrushing Test

Before starting the test, surface roughness (Ra) will be measured using a contact profilometer (MarSurf PS10). The surface of each specimen will be tested at five random different sites. The numerical roughness (Ra) values of each surface profile will be digitally calculated by the device's software. Thus, the average Ra value for each specimen will be calculated.

Artificial Toothbrushing Test

Custom Made Toothbrush Machine (Dental Biomaterial Department, Alexandria University) will be used to brush the entirety of each specimen with a vertical load of 2.0 N. Toothpaste slurry will be prepared by mixing 250 g of toothpaste (Colgate Total; relative dentin abrasion = 70) per 1 L of distilled water according to the ISO 14569-1:2007 standard; the slurry will be replaced every 5000 cycles. Each specimen will be washed by a stream of compressed tap water and dried by a disposable sponge after the cycle is completed.(11,12)

Surface Roughness Measurement after Toothbrushing Test

Each specimen will be placed again in the corresponding mold space for retesting as mentioned above.

Weighing the Sample after Toothbrushing Test

The wear resistance will be measured by weighing each specimen before and after the test and then calculating the difference in weight representing the amount of worn substance.(13)

BUDGET ESTIMATE

No	Items	Price (EGP)
1	Composites	18,000
2	Electronic Sensitive Balance	2500
3	Surface Profilometer	10,000
4	Toothbrushing Test	3500
Total		**34,000**

DURATION OF STUDY

Estimated time: 12 Months

Tasks / Months	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Sample selection/ patient rectuitment	**x**											
Clinical procedure/ Experimental procedure		**x**										
Testing and results		**x**	**x**	**x**								
Data management and statistical analysis				**x**	**x**							
Writing Thesis/manuscript					**x**	**x**	**x**	**x**				
Thesis submission									**x**	**x**	**x**	
Thesis discussion												**x**

ANTICIPATED PROBLEMS

1. Funding problems (no problems so far)

2. Availability of materials (already present)

3. Laboratory procedure problems (pilot study done)

PUBLICATION POLICY

Findings will be disseminated through peer-reviewed publication.

The order of names will be:

1. Mohamed Talaat ElAbd

2. Prof. Dr. Wegdan M. Abdel-Fattah

3. Asst. Prof. Dr. Emad M. El-Sayed

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الموافقة المستنيرة للاشتراك في دراسة علمية

اسم الباحث: محمد طلعت محمد عبد المعطي العبد
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الهدف من الدراسة: تقييم مقاومة التأكل و خشونة السطح للراتينج المركب القابل للحقن مقارنة بالراتينج المركب التقليدي

عدد الاشخاص المشاركين: ٣ أشخاص

خطوات الدراسة:

- تقدير عدد العينات
- تحضير العينات
- اختبار العينات وفقا لمعايير الإدراج
- وزن العينات
- قياس خشونة السطح
- اختبار تفريش الأسنان الاصطناعى
- وزن العينات وقياس خشونة السطح بعد التفريش
- ثم جمع البيانات وتحليها

فترة مشاركتي بالدراسة: ١٢ شهر

الموافقة المستنيرة للاشتراك في دراسة علمية (تابع)

سرية المعلومات:
سوف يتم حفظ المعلومات المتعلقة بك في سرية تامة ولن يتم التعريف بك في أي تقرير أو نشر لنتائج البحث.
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سوف يقوم الباحث باتخاذ جميع الوسائل لمنع اي اصابة أو ضرر أو أي عراض جانبية من الممكن ان يحدث نتيجة لهذه الدراسة، ولكن إذا حدثت أي اصابة غير متوقعة نتيجة اشتراكك بالدراسة سوف تتلقي العلاج اللازم بالكلية؛ ولا يوجد أي تعويض مادي عن أي إصابة قد تنتج من هذه الدراسة.
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الاشتراك تطوعي ولك الحق في ان تستمر او تنسحب من الدراسة في أي وقت دون تبعات عليك.
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لقد قمت بشرح الدراسة بالتفصيل والهدف منها وطريقة الدراسة والمخاطر والفوائد المتعلقة بالدراسة مع الاجابة عن أي أسئلة أثارها المشارك وسوف ألتزم بخطوات الدراسة كاملة وبالمعاييرالاخلاقية والقانونية للبحث العلمي بالكلية، وفي حال تغيير خطوات البحث سيكون ذلك لمصلحة المريض وبإخطار مسبق للمريض، وللمشاركين الذين لا يحسنون القراءة والكتابة تم قراءة الموافقة المستنيرة على المشارك في حضور شاهد من اختيار المشارك

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الموافقة المستنيرة للاشتراك في دراسة علمية (تابع)

موافقة المريض:

أنا الموقع ادناه قد شرح لي هدف وخطوات الدراسة ، وقد استوعبت الفوائد والمخاطر المتعلقة بالدراسة وتم
التأكيد على سرية معلوماتي الخاصة، وتم اخطاري باحتمال تغيير خطوات البحث عند الضرورة و لمصلحتي،
وقد أخذت نسخة من هذه الموافقة، وقد تم اعطائي الفرصة للسؤال والاستفسار قبل التوقيع ، وقيل لي أن من
حقي الاستفسار في اي وقت لاحق، وأنا أتطوع للمشاركة في هذه الدراسة و لي الحق في الانسحاب في اي
وقت دون تبعات علي، وأوافق علي التعاون مع الباحث و إخباره مباشرة بأي مشاكل غير متوقعة يمكن ان
تحدث لي أثناء اشتراكي في الدراسة.

اسم المشارك/ولي الأمر/ الممثل القانوني:
توقيع (ختم أو بصمة) المشارك/ولي الأمر/ الممثل القانوني:
توقيع الشاهد :
توقيع الباحث:
التاريخ :

تعهد من الباحث للجنة أخلاقيات البحوث الطبية

اسم الباحث: محمد طلعت محمد عبدالمعطي

اسم البحث باللغة العربية :

دراسة مقارنة للتآكل و خشونة سطح الراتينج المركب القابل للحقن و التقليدي ـ دراسة في المختبر

اسم البحث باللغة الإنجليزية :

Comparative Study of Wear Resistance and Surface Roughness of Injectable Versus Conventional Composite Resin - In Vitro Study

القسم العلمي: العلاج التحفظي

عنوان الباحث: ١١ ش محمود حسن فهمي, الابراهيمية, الاسكندرية

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أتعهد أنا الموقع أدناه بمسئوليتي الكاملة عن استيفاء نموذج الموافقة المستنيرة المرفق بخطة البحث لكل مشارك قبل الخضوع للبحث، مع الاحتفاظ بأصل الموافقة المستنيرة وتسليم المشارك نسخة موقعة، ومن حق اللجنة الاطلاع عليها في أي وقت.
كما أؤكد أنه تم منح المشارك فرصة لطرح أسئلة حول الدراسة، وتم الإجابة على جميع الأسئلة التي طرحها المشارك بشكل صحيح وبقدر المستطاع ، وأؤكد أن أي فرد لم يُجبر على إعطاء الموافقة، وأن الموافقة تم منحها طواعيه وبدون إكراه.

توقيع الباحث:
توقيع المشرف الرئيسي:
التاريخ